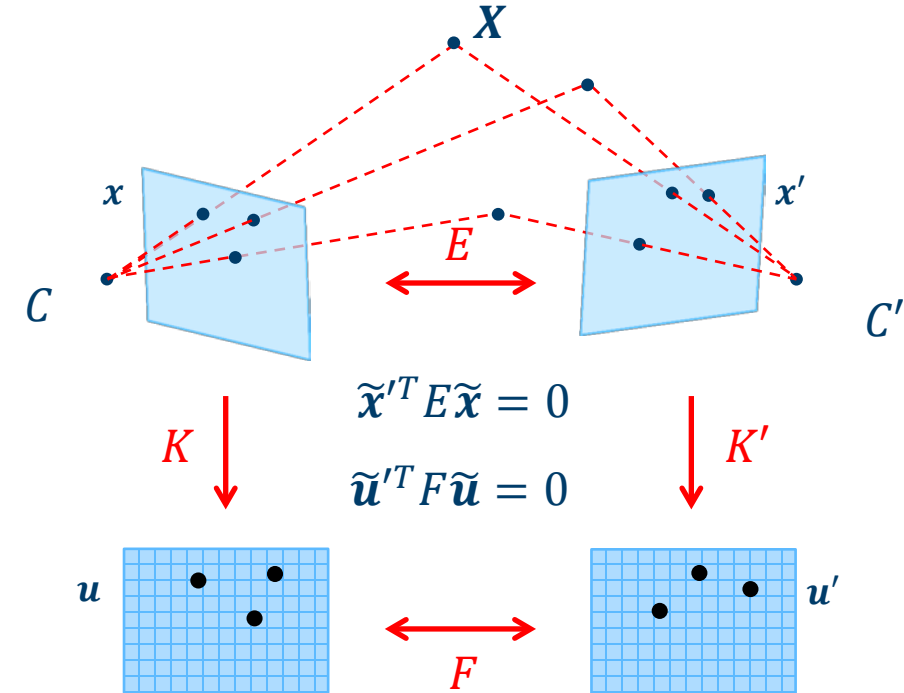


Lecture 7.2

Triangulation

Introduction

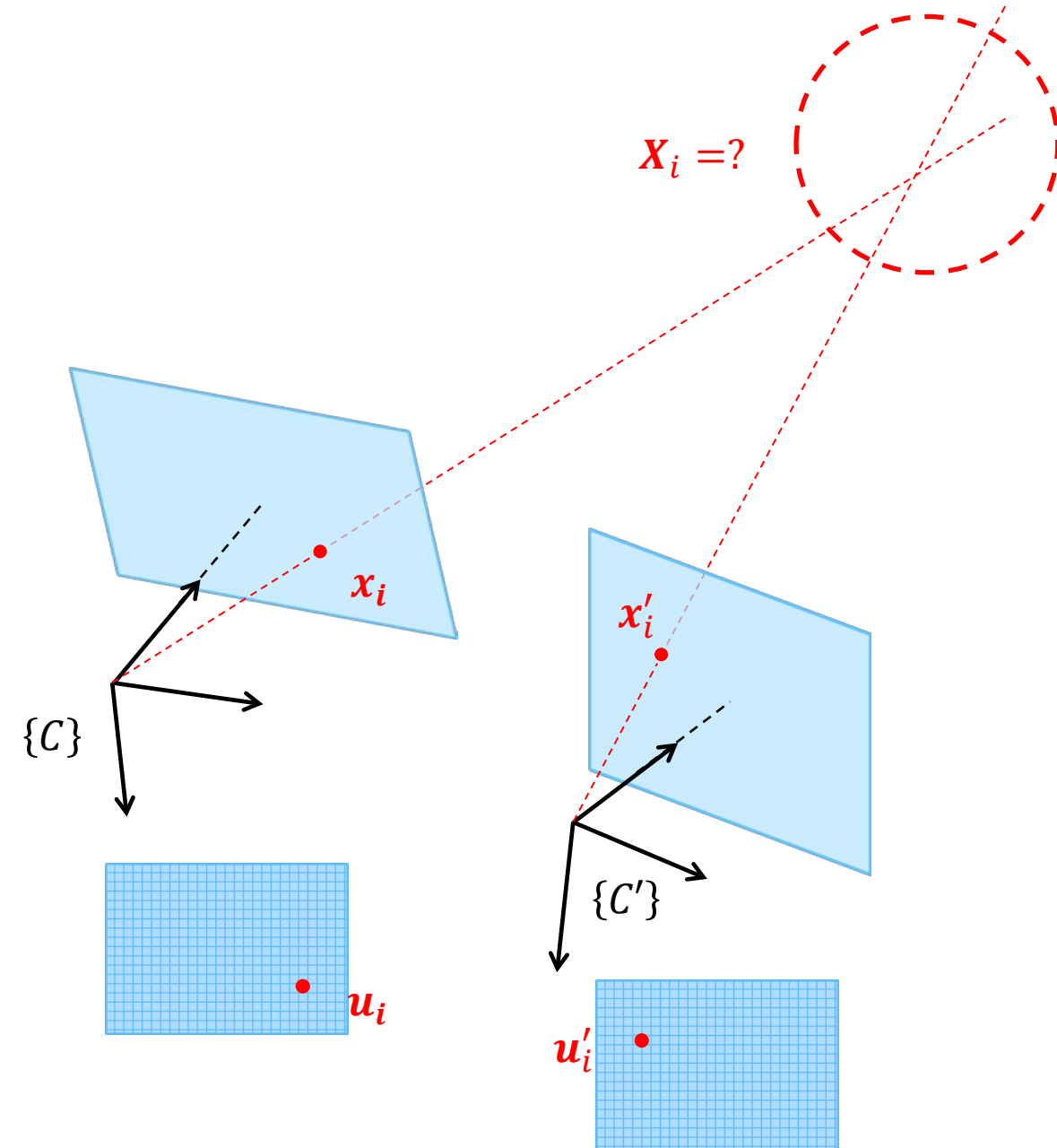
- We have seen that two perspective cameras observing the same points must satisfy the epipolar constraint
- The essential matrix $E = [t]_{\times} R$
 $\tilde{x}'^T E \tilde{x} = 0$
- The fundamental matrix $F = K'^{-T} E K^{-1}$
 $\tilde{u}'^T F \tilde{u} = 0$
- Being observed by two perspective cameras also puts a strong geometric constraint on the observed points X_i
 $PX_i = u_i$
 $P'X_i = u'_i$



- In the following we will look at how we can estimate 3D points X_i from known camera matrices P , P' and 2D correspondences $u_i \leftrightarrow u'_i$

Introduction

- Assume that we know the camera matrices P , P' and 2D correspondences $\mathbf{u}_i \leftrightarrow \mathbf{u}'_i$
- In order to determine the 3D point $\mathbf{X}_i$ it is tempting to back-project the two image points and determine their intersection
- But **due to noise**, the two rays in 3D will “never” truly intersect, so we need to estimate a best solution to the problem
- Several ways to approach the problem depending on what we choose to optimize over
 - Only errors in $\mathbf{u}_i \leftrightarrow \mathbf{u}'_i$?
 - Errors in $\mathbf{u}_i \leftrightarrow \mathbf{u}'_i$, P and P' ?



The 3D mid-point

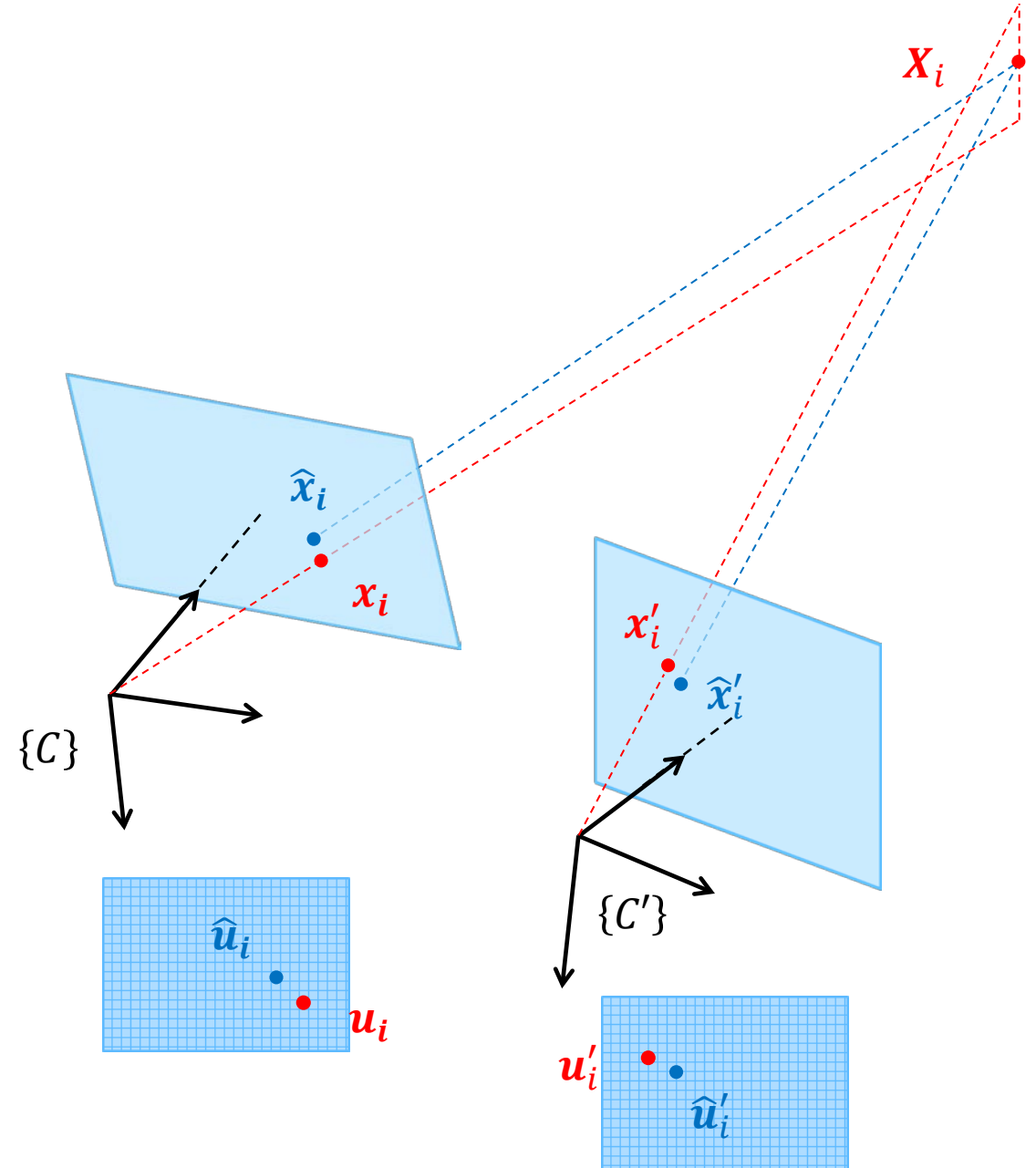
Minimizing the 3D error

- One natural estimate for X_i is the midpoint on the shortest line between two back-projected rays

- This minimize the 3D error, but typically not the reprojection error

$$\epsilon_i = d(\mathbf{u}_i, P\mathbf{X}_i)^2 + d(\mathbf{u}'_i, P'\mathbf{X}_i)^2$$

$$\epsilon_i = d(\mathbf{u}_i, \hat{\mathbf{u}}_i)^2 + d(\mathbf{u}'_i, \hat{\mathbf{u}}'_i)^2$$



The 3D mid-point

Minimizing the 3D error

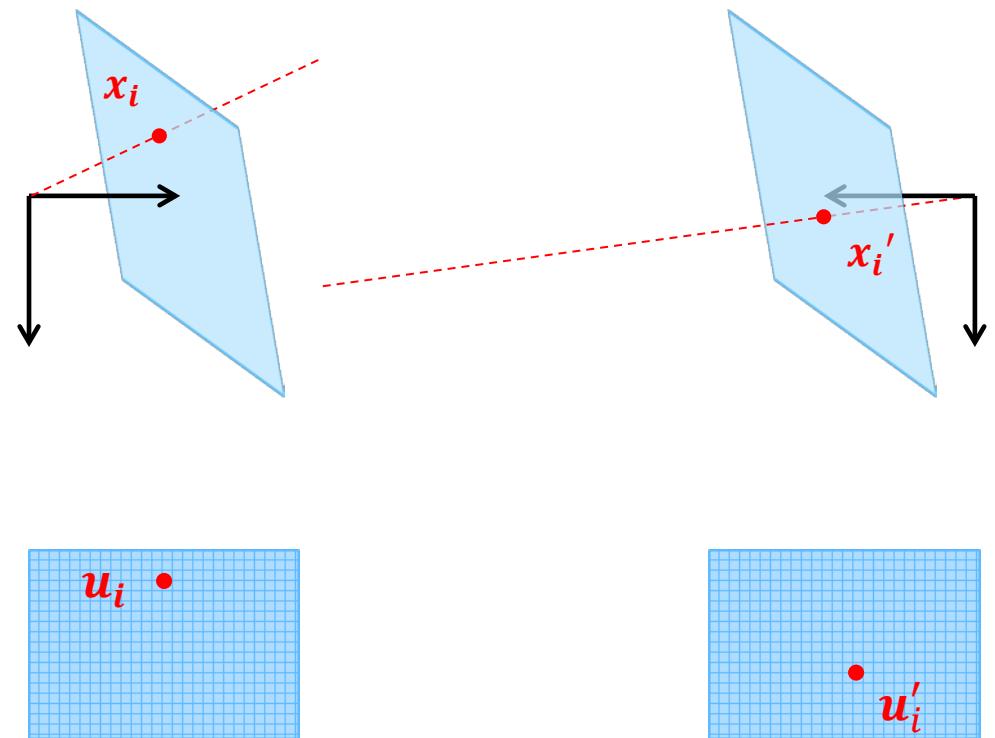
- One natural estimate for $\mathbf{X}_i$ is the midpoint on the shortest line between two back-projected rays

- This minimize the 3D error, but typically not the reprojection error

$$\epsilon_i = d(\mathbf{u}_i, P\mathbf{X}_i)^2 + d(\mathbf{u}'_i, P'\mathbf{X}_i)^2$$

$$\epsilon_i = d(\mathbf{u}_i, \hat{\mathbf{u}}_i)^2 + d(\mathbf{u}'_i, \hat{\mathbf{u}}'_i)^2$$

- The difference becomes clear when $\mathbf{X}_i$ is much closer to one of the cameras than the other



The 3D mid-point

Minimizing the 3D error

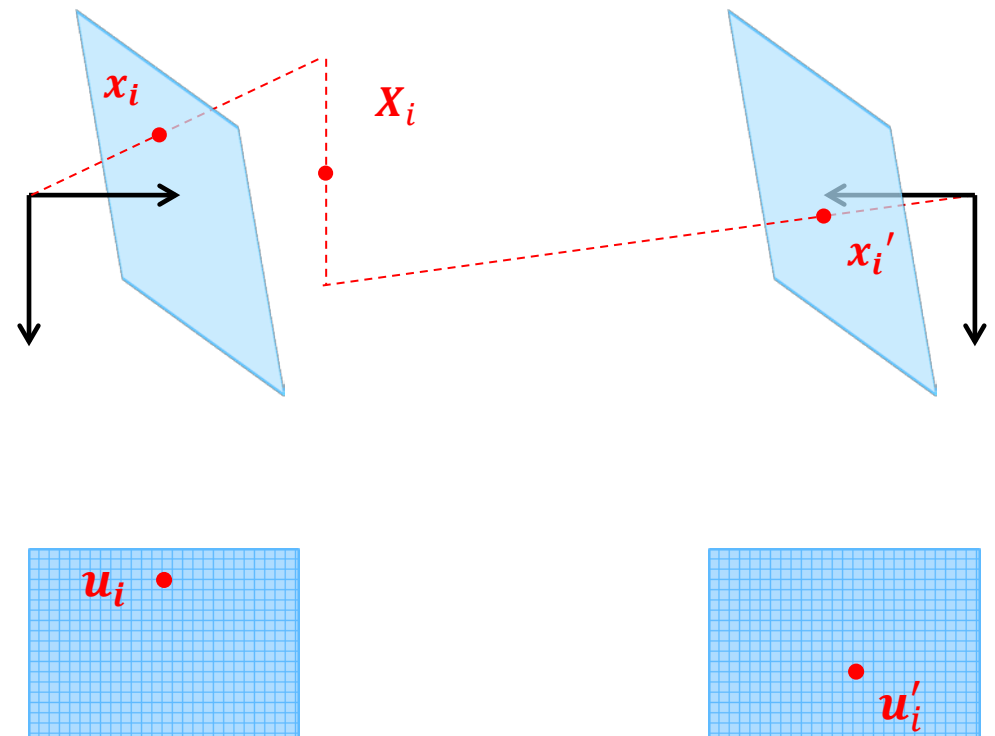
- One natural estimate for $\mathbf{X}_i$ is the midpoint on the shortest line between two back-projected rays

- This minimize the 3D error, but typically not the reprojection error

$$\epsilon_i = d(\mathbf{u}_i, P\mathbf{X}_i)^2 + d(\mathbf{u}'_i, P'\mathbf{X}_i)^2$$

$$\epsilon_i = d(\mathbf{u}_i, \hat{\mathbf{u}}_i)^2 + d(\mathbf{u}'_i, \hat{\mathbf{u}}'_i)^2$$

- The difference becomes clear when $\mathbf{X}_i$ is much closer to one of the cameras than the other



The 3D mid-point

Minimizing the 3D error

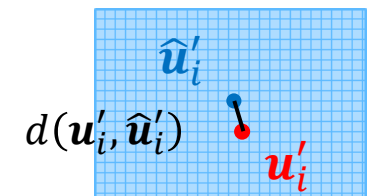
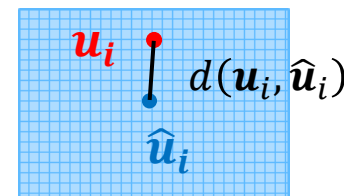
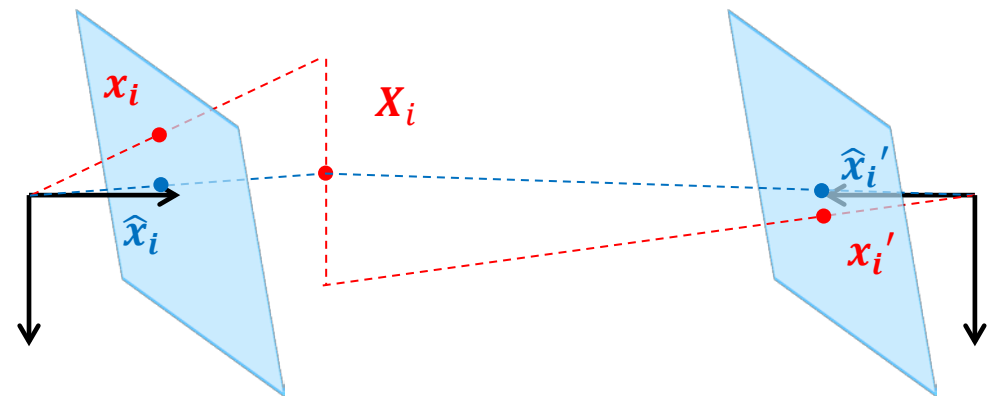
- One natural estimate for $\mathbf{X}_i$ is the midpoint on the shortest line between two back-projected rays

- This minimizes the 3D error, but typically not the reprojection error

$$\epsilon_i = d(\mathbf{u}_i, P\mathbf{X}_i)^2 + d(\mathbf{u}'_i, P'\mathbf{X}_i)^2$$

$$\epsilon_i = d(\mathbf{u}_i, \hat{\mathbf{u}}_i)^2 + d(\mathbf{u}'_i, \hat{\mathbf{u}}'_i)^2$$

- The difference becomes clear when $\mathbf{X}_i$ is much closer to one of the cameras than the other



The 3D mid-point

Minimizing the 3D error

- One natural estimate for X_i is the midpoint on the shortest line between two back-projected rays

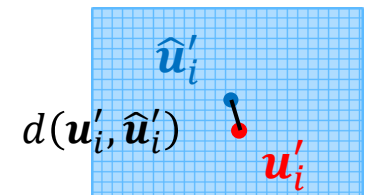
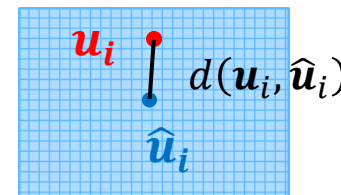
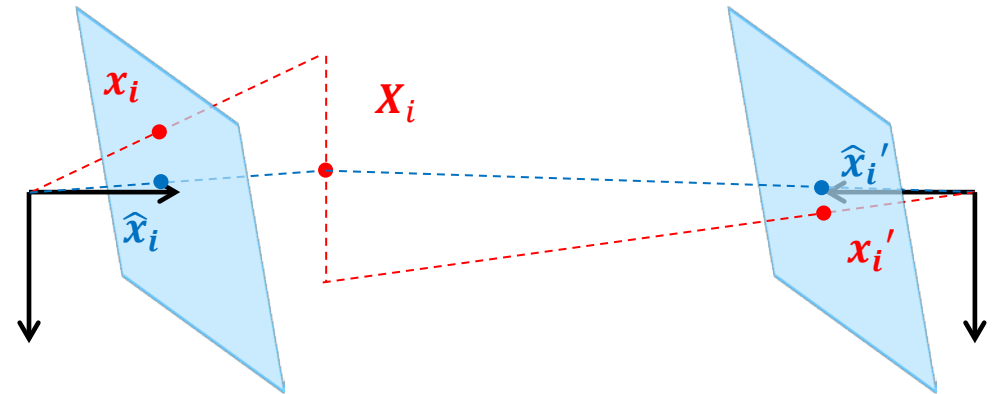
- This minimizes the 3D error, but typically not the reprojection error

$$\epsilon_i = d(\mathbf{u}_i, P\mathbf{X}_i)^2 + d(\mathbf{u}'_i, P'\mathbf{X}_i)^2$$

$$\epsilon_i = d(\mathbf{u}_i, \hat{\mathbf{u}}_i)^2 + d(\mathbf{u}'_i, \hat{\mathbf{u}}'_i)^2$$

- The difference becomes clear when X_i is much closer to one of the cameras than the other

- Another disadvantage of this method is that it does not extend naturally to situations when X_i is observed by more than two cameras



Linear triangulation

Minimizing the algebraic error

- This algorithm uses the two equations for perspective projection to solve for the 3D point that are optimal in a least squares sense
- Each perspective camera model gives rise to two equations on the three entries of $\mathbf{X}_i$

$$\begin{aligned}
 \tilde{\mathbf{u}}_i &= P\tilde{\mathbf{X}}_i \\
 \Downarrow \\
 \tilde{\mathbf{u}}_i \times P\tilde{\mathbf{X}}_i &= \mathbf{0} \\
 \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \times \begin{bmatrix} \mathbf{p}^{1T} \\ \mathbf{p}^{2T} \\ \mathbf{p}^{3T} \end{bmatrix} \tilde{\mathbf{X}}_i &= \mathbf{0} \\
 \begin{bmatrix} v_i \mathbf{p}^{3T} - \mathbf{p}^{2T} \\ \mathbf{p}^{1T} - u_i \mathbf{p}^{3T} \\ u_i \mathbf{p}^{2T} - v_i \mathbf{p}^{1T} \end{bmatrix} \tilde{\mathbf{X}}_i &= \mathbf{0} \\
 \Updownarrow \\
 \begin{bmatrix} v_i \mathbf{p}^{3T} - \mathbf{p}^{2T} \\ u_i \mathbf{p}^{3T} - \mathbf{p}^{1T} \end{bmatrix} \tilde{\mathbf{X}}_i &= \mathbf{0}
 \end{aligned}$$

$$\begin{aligned}
 \tilde{\mathbf{u}}'_i &= P'\tilde{\mathbf{X}}_i \\
 \Downarrow \\
 \tilde{\mathbf{u}}'_i \times P'\tilde{\mathbf{X}}_i &= \mathbf{0} \\
 \begin{bmatrix} u'_i \\ v'_i \\ 1 \end{bmatrix} \times \begin{bmatrix} \mathbf{p}'^{1T} \\ \mathbf{p}'^{2T} \\ \mathbf{p}'^{3T} \end{bmatrix} \tilde{\mathbf{X}}_i &= \mathbf{0} \\
 \begin{bmatrix} v'_i \mathbf{p}'^{3T} - \mathbf{p}'^{2T} \\ \mathbf{p}'^{1T} - u'_i \mathbf{p}'^{3T} \\ u'_i \mathbf{p}'^{2T} - v'_i \mathbf{p}'^{1T} \end{bmatrix} \tilde{\mathbf{X}}_i &= \mathbf{0} \\
 \Updownarrow \\
 \begin{bmatrix} v'_i \mathbf{p}'^{3T} - \mathbf{p}'^{2T} \\ u'_i \mathbf{p}'^{3T} - \mathbf{p}'^{1T} \end{bmatrix} \tilde{\mathbf{X}}_i &= \mathbf{0}
 \end{aligned}$$

Linear triangulation

Minimizing the algebraic error

- This algorithm uses the two equations for perspective projection to solve for the 3D point that are optimal in a least squares sense
- Each perspective camera model gives rise to two equations on the three entries of $\mathbf{X}_i$
- Combining these equations we get an over determined homogeneous system of linear equations that we can solve with SVD

$$\begin{bmatrix} v_i \mathbf{p}^{3T} - \mathbf{p}^{2T} \\ u_i \mathbf{p}^{3T} - \mathbf{p}^{1T} \\ v'_i \mathbf{p}'^{3T} - \mathbf{p}'^{2T} \\ u'_i \mathbf{p}'^{3T} - \mathbf{p}'^{1T} \end{bmatrix} \tilde{\mathbf{X}}_i = \mathbf{0}$$

$$\begin{bmatrix} v_i p_{31} - p_{21} & v_i p_{32} - p_{22} & v_i p_{33} - p_{23} & v_i p_{34} - p_{24} \\ u_i p_{31} - p_{11} & u_i p_{32} - p_{12} & u_i p_{33} - p_{13} & u_i p_{34} - p_{14} \\ v'_i p'_{31} - p'_{21} & v'_i p'_{32} - p'_{22} & v'_i p'_{33} - p'_{23} & v'_i p'_{34} - p'_{24} \\ u'_i p'_{31} - p'_{11} & u'_i p'_{32} - p'_{12} & u'_i p'_{33} - p'_{13} & u'_i p'_{34} - p'_{14} \end{bmatrix} \tilde{\mathbf{X}}_i = \mathbf{0}$$

$$A \tilde{\mathbf{X}}_i = \mathbf{0}$$

Linear triangulation

Minimizing the algebraic error

- This algorithm uses the two equations for perspective projection to solve for the 3D point that are optimal in a least squares sense
- Each perspective camera model gives rise to two equations on the three entries of $\mathbf{X}_i$
- Combining these equations we get an over determined homogeneous system of linear equations that we can solve with SVD
- The minimized algebraic error is not geometrically meaningful, but the method extends naturally to the case when $\mathbf{X}_i$ is observed in more than two images

$$\begin{bmatrix} v_i \mathbf{p}^{3T} - \mathbf{p}^{2T} \\ u_i \mathbf{p}^{3T} - \mathbf{p}^{1T} \\ v'_i \mathbf{p}'^{3T} - \mathbf{p}'^{2T} \\ u'_i \mathbf{p}'^{3T} - \mathbf{p}'^{1T} \\ \vdots \end{bmatrix} \tilde{\mathbf{X}}_i = \mathbf{0}$$

$$\begin{bmatrix} v_i p_{31} - p_{21} & v_i p_{32} - p_{22} & v_i p_{33} - p_{23} & v_i p_{34} - p_{24} \\ u_i p_{31} - p_{11} & u_i p_{32} - p_{12} & u_i p_{33} - p_{13} & u_i p_{34} - p_{14} \\ v'_i p'_{31} - p'_{21} & v'_i p'_{32} - p'_{22} & v'_i p'_{33} - p'_{23} & v'_i p'_{34} - p'_{24} \\ u'_i p'_{31} - p'_{11} & u'_i p'_{32} - p'_{12} & u'_i p'_{33} - p'_{13} & u'_i p'_{34} - p'_{14} \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \tilde{\mathbf{X}}_i = \mathbf{0}$$

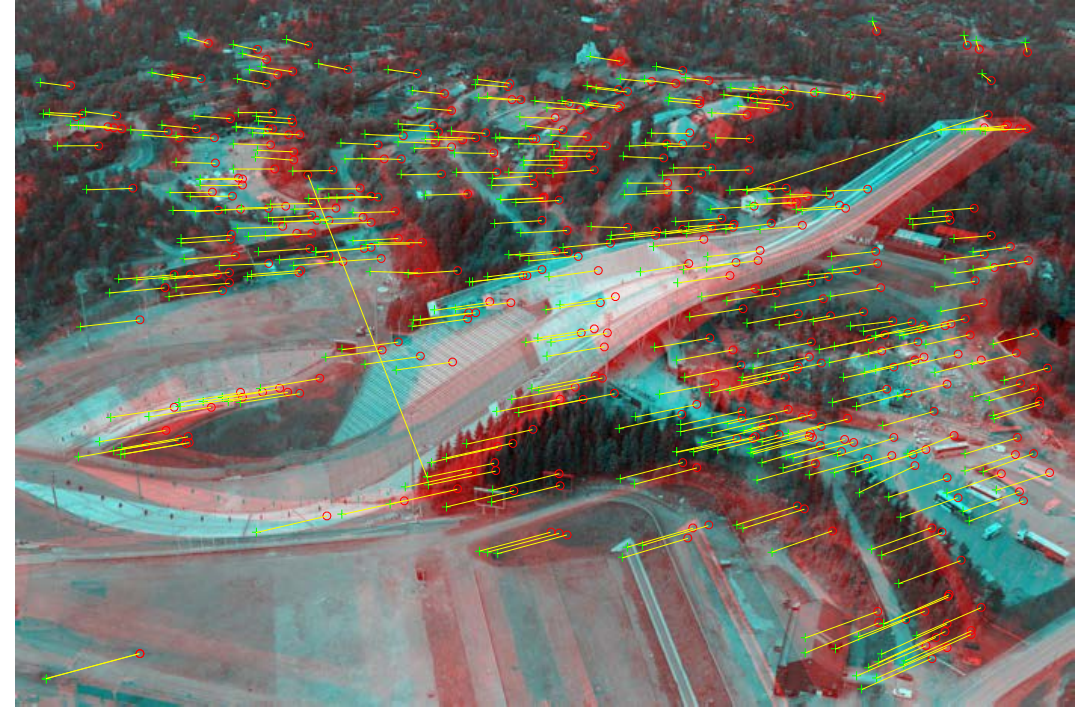
$$A \tilde{\mathbf{X}}_i = \mathbf{0}$$

Example



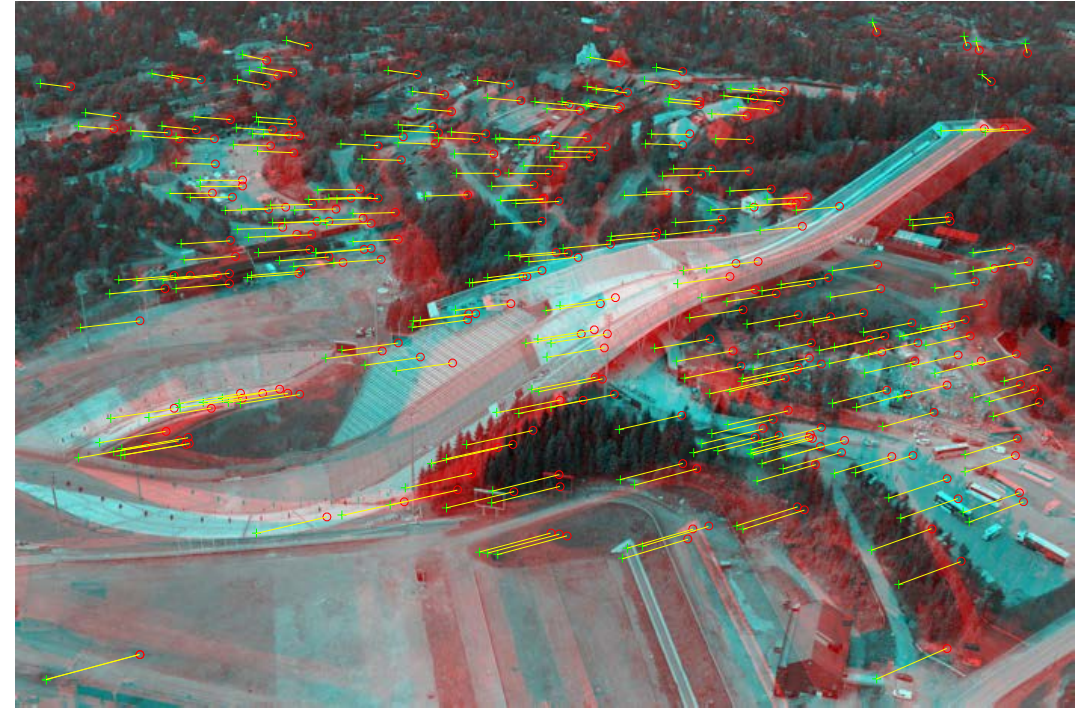
- Two views with known relative pose

Example



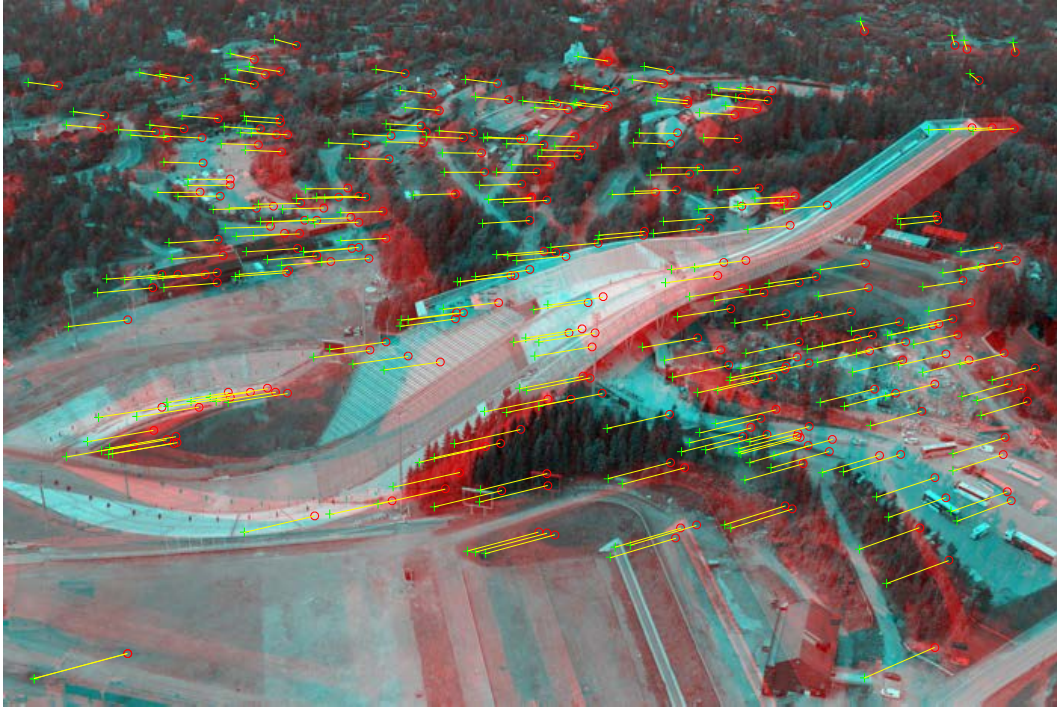
- Two views with known relative pose
- Matching feature points

Example

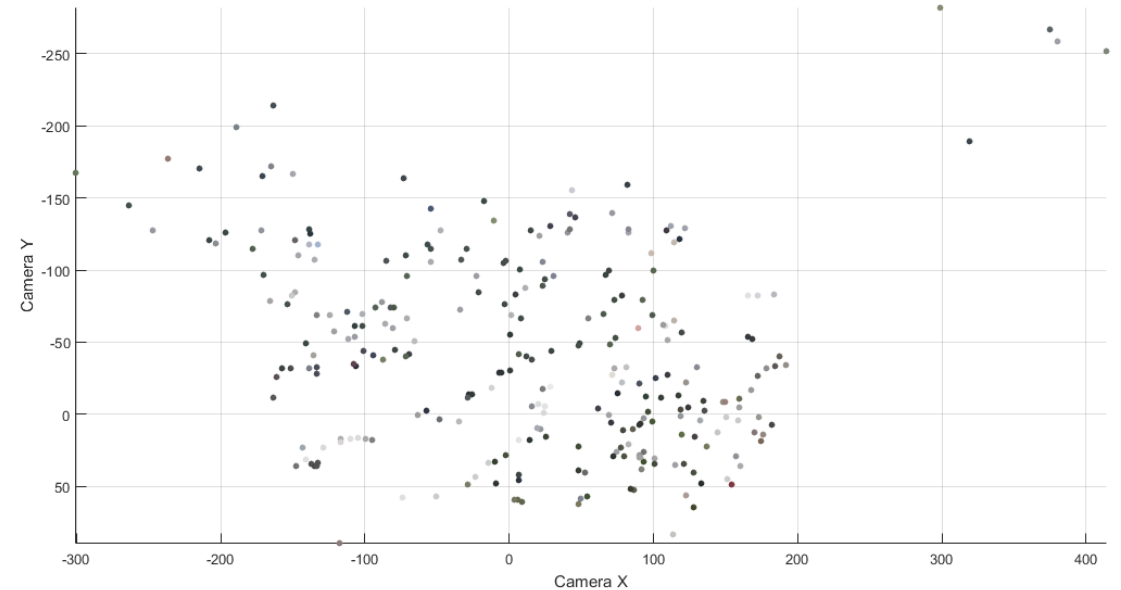


- Two views with known relative pose
- Matching feature points
- After filtering by the epipolar constraint
 - Keeping matches that are within ± 0.5 pixels of the epipolar line

Example

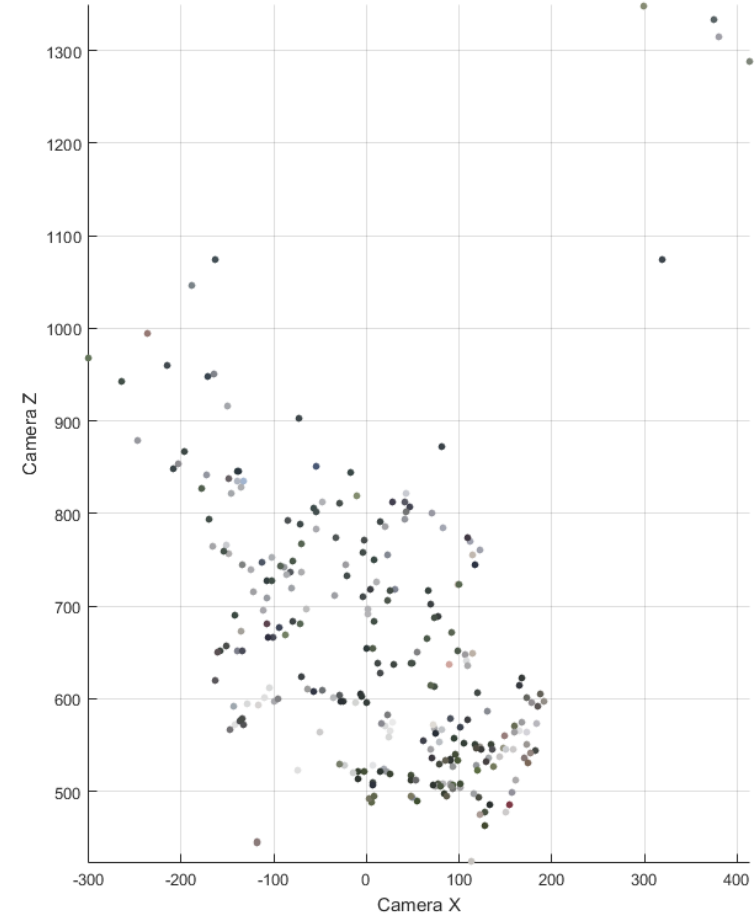
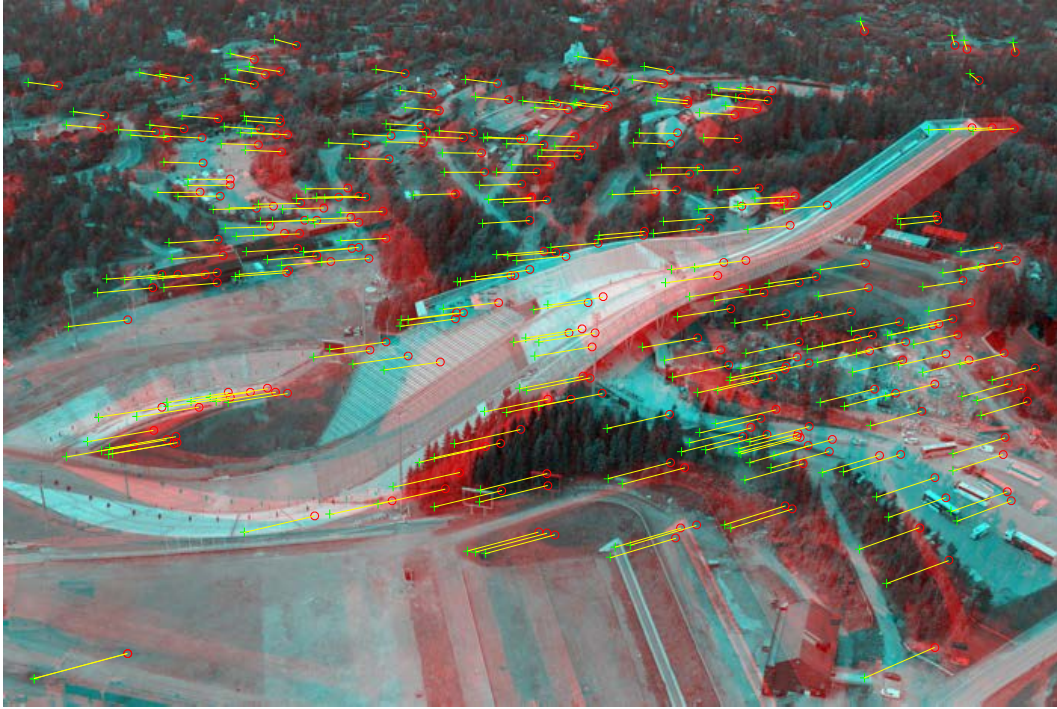


- Two views with known relative pose
- Matching feature points
- After filtering by the epipolar constraint
 - Keeping matches that are within ± 0.5 pixels of the epipolar line



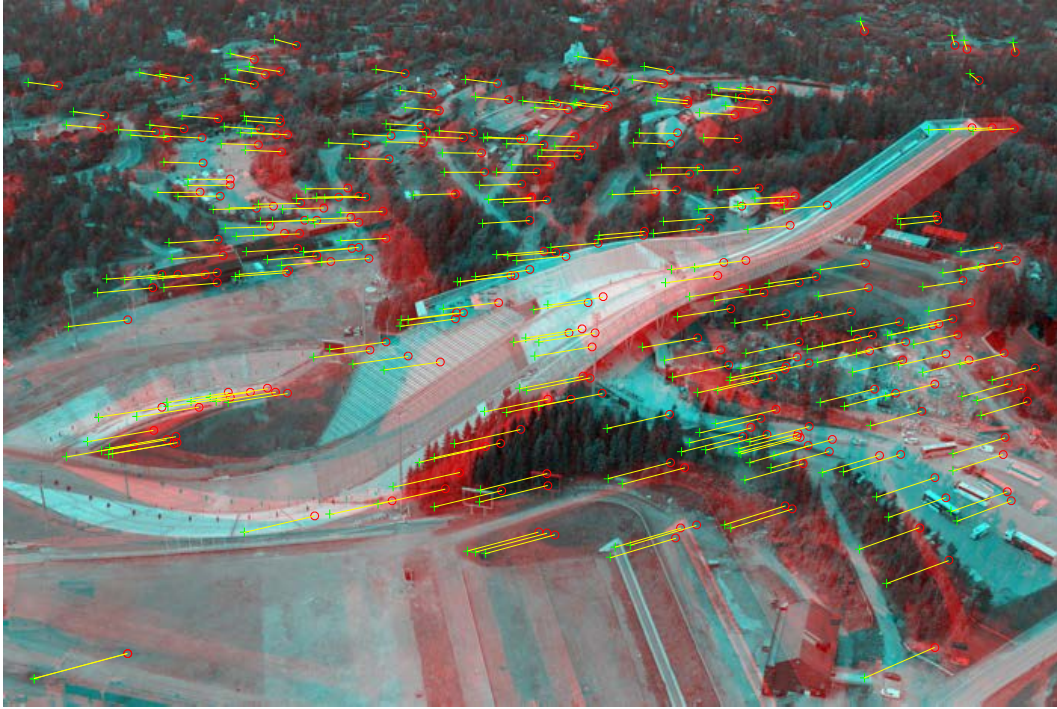
- Sparse 3D reconstruction of the scene by triangulation

Example

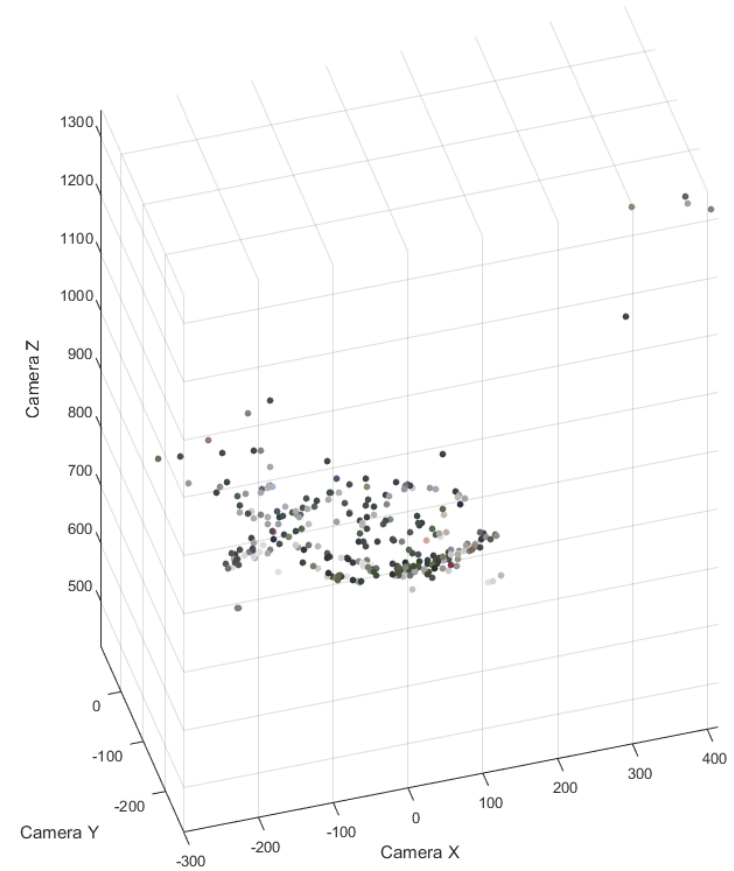


- Two views with known relative pose
- Matching feature points
- After filtering by the epipolar constraint
 - Keeping matches that are within ± 0.5 pixels of the epipolar line
- Sparse 3D reconstruction of the scene by triangulation

Example



- Two views with known relative pose
- Matching feature points
- After filtering by the epipolar constraint
 - Keeping matches that are within ± 0.5 pixels of the epipolar line



- Sparse 3D reconstruction of the scene by triangulation

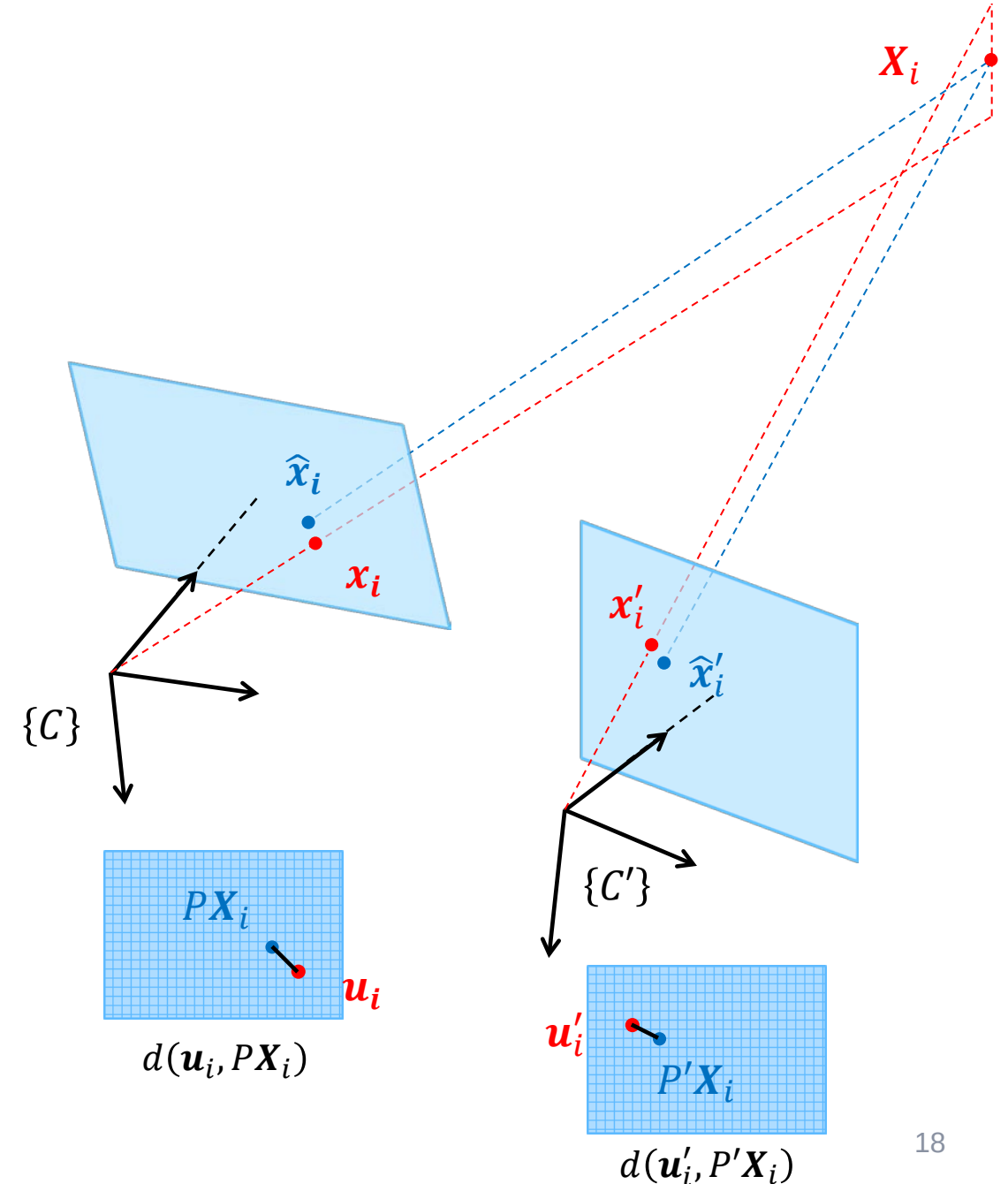
Non-linear triangulation

Minimizing a geometric error

- Compared to the previous algorithms it would be better to find the 3d point X_i that minimize a meaningful geometric error, like the reprojection error

$$\epsilon_i = d(\mathbf{u}_i, P\mathbf{X}_i)^2 + d(\mathbf{u}'_i, P'\mathbf{X}_i)^2$$

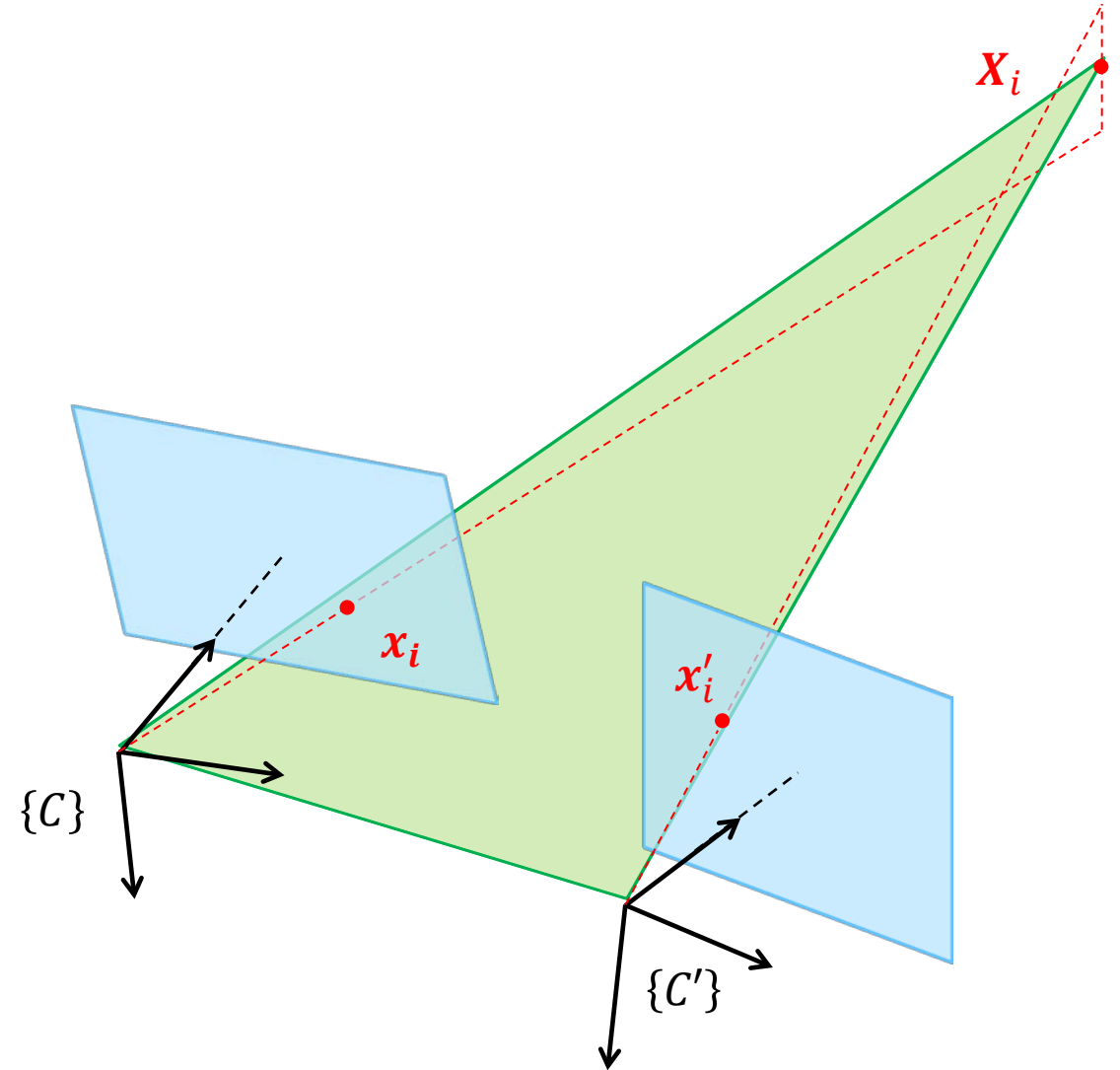
- It can be shown that if the measurement noise in image points is Gaussian with mean equal to zero, the minimizing the reprojection error gives the Maximum Likelihood estimate of X_i
- At first glance, this minimization appears to be over the three parameters in X_i , but under the assumption that P and P' are error free the problem can be reduced to a minimization over one parameter



Non-linear triangulation

Minimizing a geometric error

- If P and P' are error free, then the epipolar geometry is error free
 - We have a unique baseline which define all possible epipolar planes as a 1-parameter family
 - We have unique epipoles that all epipolar lines must pass through, so we have 1-parameter families of epipolar lines as well
- By requiring that both reprojected points lie on the same epipolar plane, the minimization problem can be reformulated in terms of the 1-parameter families of epipolar lines



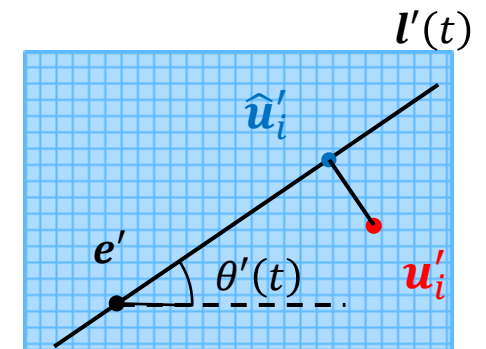
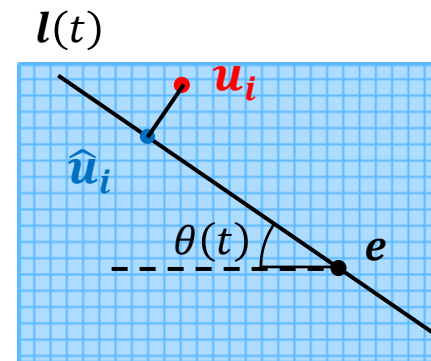
Non-linear triangulation

Minimizing a geometric error

- If P and P' are error free, then the epipolar geometry is error free
 - We have a unique baseline which define all possible epipolar planes as a 1-parameter family
 - We have unique epipoles that all epipolar lines must pass through, so we have 1-parameter families of epipolar lines as well
- By requiring that both reprojected points lie on the same epipolar plane, the minimization problem can be reformulated in terms of the 1-parameter families of epipolar lines

$$\epsilon_i = d(\mathbf{u}_i, \mathbf{l}(t))^2 + d(\mathbf{u}'_i, \mathbf{l}'(t))^2$$

- To find the t that minimize the reprojection error one has to minimize a 6th degree polynomial in t



- More details about this method and comparison with other methods can be found in the 1997 paper *Triangulation* by R. I. Hartley and P. Sturm

Summary

- **Triangulation** – Estimate a 3D point X_i for a noisy 2D correspondence under the assumption that camera matrices P and P' are known
- **Minimal 3D error** – Choose X_i to be the mid-point between back projected image points
- **Minimal algebraic error** – Combine the two perspective models to get a homogeneous system of linear equations, then determine X_i by SVD
- **Minimal reprojection error** – Determine the epipolar plane (and points $\hat{u}_i$ and $\hat{u}'_i$) that minimize the reprojection error by minimizing a 6th order polynomial
- Additional reading
 - Szeliski: 7.1
- Optional reading
 - R. I. Hartley and P. Sturm, *Triangulation* (1997)

